

PADHAISHURU

GATE Preparation Platform · padhaishuru.com

TREND-BASED MOCK PAPER

GATE Geology and Geophysics

Mock Paper — 02

Full-Length Examination Simulation

Duration: 3 Hours | Maximum Marks: 100 | Questions: 65

Examination Details

Branch:	Geology and Geophysics
Paper:	Mock 02
Total Questions:	65
Total Marks:	100
Duration:	3 Hours
Question Types:	MCQ, MSQ, NAT
Negative Marking:	Yes (MCQ only)
Difficulty:	Mixed

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.
Historical patterns do not guarantee future examination questions.
This is a trend-based practice simulation.

GENERAL INSTRUCTIONS

1. This question paper contains 65 questions for a total of 100 marks.
2. Duration of examination: 3 Hours.
3. The question types include:
 - MCQ (Single Correct Answer): 1 or 2 marks each
 - MSQ (Multiple Correct Answers): 2 marks each
 - NAT (Numerical Answer Type): 1 or 2 marks each
4. For MCQ: Only one option is correct. Select the single correct answer.
5. For MSQ: One or more options are correct. Select ALL correct options.
6. For NAT: Enter the numerical answer. No negative marking for NAT questions.
7. Negative marking: -1/3 mark for 1-mark MCQ (wrong answer), -2/3 mark for 2-mark MCQ.
No negative marking for MSQ and NAT questions.
8. A virtual calculator is available on the screen. Physical calculators are not permitted.
9. The question paper contains 8 pages including this instructions page.
10. All questions are compulsory.
11. Fill in the required details on the answer sheet before starting.
12. Do not ask for clarifications from the invigilator. Interpret questions as given.

Unofficial Practice Material — Not affiliated with or endorsed by GATE/IIT.

Prepared by PadhaiShuru (padhaishuru.com) for practice purposes only.

Section A: General Aptitude

Q1. [1-NAT] [1 mark]

Choose similar to 'volcanic':

(Numerical Answer Type — Enter the numerical value)

Section B: Geology and Geophysics

Q2. [1-NAT] [1 mark]

Mean of 1,2,3,4,5:

(Numerical Answer Type — Enter the numerical value)

Q3. [1-MCQ] [1 mark]

For matrix $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$, $A^{50} \cdot [1, 0]^T$ equals:

- (A) $[1, 0]^T$
- (B) $[2, 1]^T$
- (C) $[(3+2\sqrt{2})^{50}, \dots]$
- (D) Cannot determine

Q4. [1-MCQ] [1 mark]

$\frac{d}{dx}[\ln(\sec x) + \ln(\tan x)]$ evaluated at $x = \pi/4$ equals:

- (A) 0
- (B) 2
- (C) 4
- (D) 1

Q5. [1-MCQ] [1 mark]

$\lim_{n \rightarrow \infty} (1 + 1/n)^{2n}$ equals:

- (A) e

- (B) e^2
- (C) 1
- (D) infinity

Q6. [1-MCQ] [1 mark]

Laplace transform of $e^{(-3t)}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) e^{-3}/s

Q7. [1-NAT] [1 mark]

The differential equation $y \, dx - x \, dy = 0$ has solution:

(Numerical Answer Type — Enter the numerical value)

Q8. [1-MCQ] [1 mark]

$P(|Z| \leq 1)$ for $Z \sim N(0,1)$ is approximately:

- (A) 0.68
- (B) 0.32
- (C) 0.95
- (D) 0.16

Q9. [1-NAT] [1 mark]

Number of spanning trees in complete graph K_4 :

(Numerical Answer Type — Enter the numerical value)

Q10. [1-NAT] [1 mark]

The vector $(1,2,3)$ projected onto $(2,1,0)$ is:

(Numerical Answer Type — Enter the numerical value)

Q11. [1-MCQ] [1 mark]

If $f(x) = x^3 - 3x + 2$, number of real roots:

- (A) 1
- (B) 2
- (C) 3
- (D) 0

Q12. [1-NAT] [1 mark]

The rank of the matrix $\begin{bmatrix} 1,2,3 \\ 2,4,6 \\ 3,6,9 \end{bmatrix}$ is:

(Numerical Answer Type — Enter the numerical value)

Q13. [1-MCQ] [1 mark]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q14. [1-MCQ] [1 mark]

The Laplace transform of $t^2 e^{-2t}$ is:

- (A) $1/(s+2)^2$
- (B) $1/(s-2)^2$
- (C) $s/(s+2)^2$
- (D) $2/(s+2)^2$

Q15. [1-MCQ] [1 mark]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite
- (C) Diagonally dominant
- (D) Non-singular

Q16. [1-MCQ] [1 mark]

Consider $f(x) = |x|$ on $[-1, 1]$. This function is:

- (A) Differentiable everywhere
- (B) Continuous but not differentiable at 0
- (C) Neither continuous nor differentiable
- (D) Differentiable at 0 only

Q17. [1-NAT] [1 mark]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

(Numerical Answer Type — Enter the numerical value)

Q18. [1-NAT] [1 mark]

Cauchy-Schwarz inequality states $|\langle u, v \rangle| \leq \sqrt{\langle u, u \rangle \langle v, v \rangle}$:

(Numerical Answer Type — Enter the numerical value)

Q19. [1-MCQ] [1 mark]

The area of a circle of radius r is:

- (A) πr^2
- (B) $2\pi r$
- (C) πr
- (D) $2\pi r^2$

Q20. [1-MCQ] [1 mark]

For the ODE $dy/dx = y/x + \sqrt{y^2/x^2}$, with $y(1)=0$, the solution is:

- (A) $y = 0$
- (B) $y = x \ln(x)$
- (C) $y = x \sin(\ln x)$
- (D) $y = x^2$

Q21. [2-NAT] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\text{Re}(f(i)) =$

(Numerical Answer Type — Enter the numerical value)

Q22. [2-MCQ] [2 marks]

$P(A \cup B) = 0.6$, $P(A \cap B) = 0.2$, $P(B) = 0.4$. Then $P(A|B)$ equals:

- (A) 0.2

- (B) 0.5
- (C) 0.6
- (D) 0.4

Q23. [2-NAT] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

(Numerical Answer Type — Enter the numerical value)

Q24. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q25. [2-MCQ] [2 marks]

Which of the following is a probability generating function for $\text{Poisson}(\lambda)$?

- (A) $e^{\lambda(z-1)}$
- (B) $1/(1-z)$
- (C) $\ln(1-z)$
- (D) $e^{-\lambda}(1+z)$

Q26. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q27. [2-MSQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q28. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q29. [2-MCQ] [2 marks]

Number of spanning trees in complete graph K_4 :

- (A) 4
- (B) 8
- (C) 16
- (D) 12

Q30. [2-MCQ] [2 marks]

Gauss-Seidel method converges when the coefficient matrix is:

- (A) Symmetric
- (B) Positive definite

- (C) Diagonally dominant
- (D) Non-singular

Q31. [2-NAT] [2 marks]

For a Poisson distribution with mean 4, variance is:

(Numerical Answer Type — Enter the numerical value)

Q32. [2-MCQ] [2 marks]

If $f(z) = z^2 + 3z + 2$ is analytic, then $\operatorname{Re}(f(i)) =$

- (A) -1
- (B) -2
- (C) 1
- (D) 2

Q33. [2-MCQ] [2 marks]

A random variable $X \sim \text{Poisson}(\lambda=4)$. $P(X \geq 2)$ is approximately:

- (A) 0.908
- (B) 0.092
- (C) 0.762
- (D) 0.195

Q34. [2-MCQ] [2 marks]

For a Poisson distribution with mean 4, variance is:

- (A) 4
- (B) 2
- (C) 16
- (D) 1

Q35. [2-MCQ] [2 marks]

Rolle's theorem requires $f(x)$ to be:

- (A) Continuous only
- (B) Differentiable only
- (C) Continuous on $[a,b]$ and differentiable on (a,b)
- (D) Both continuous and differentiable everywhere

Q36. [2-MCQ] [2 marks]

Laplace transform of $e^{-3t}u(t)$ is:

- (A) $1/(s+3)$
- (B) $1/(s-3)$
- (C) $1/s$
- (D) $e^{-3/s}$

Q37. [2-NAT] [2 marks]

The Laplace transform of $t e^{-2t}$ is:

(Numerical Answer Type — Enter the numerical value)

Q38. [2-MCQ] [2 marks]

The rank of the matrix $\begin{bmatrix} 1, 2, 3 \\ 2, 4, 6 \\ 3, 6, 9 \end{bmatrix}$ is:

- (A) 1
- (B) 2
- (C) 3

(D) 0

Q39. [2-MCQ] [2 marks]

Fourier series of $f(x)=x$ on $(-\pi,\pi)$ has only:

- (A) Sine terms
- (B) Cosine terms
- (C) Both
- (D) Neither

Q40. [2-MCQ] [2 marks]

If a 3×3 matrix has eigenvalues 1, 2, and 3, then $\text{trace}(A^2)$ equals:

- (A) 14
- (B) 6
- (C) 36
- (D) 12

Q41. [2-NAT] [2 marks]

Rolle's theorem requires $f(x)$ to be:

(Numerical Answer Type — Enter the numerical value)

Q42. [2-MCQ] [2 marks]

Solution of $y'' + 4y = 0$ with $y(0)=1$, $y'(0)=0$, $y(\pi/2)$ equals:

- (A) 1
- (B) 0
- (C) -1
- (D) $\cos(\pi)$

Q43. [2-NAT] [2 marks]

The area of a circle of radius r is:

(Numerical Answer Type — Enter the numerical value)

Q44. [1-MCQ] [1 mark]

Study of fossils:

- (A) Paleontology
- (B) Petrology
- (C) Mineralogy
- (D) None

Q45. [1-MCQ] [1 mark]

Hardest mineral:

- (A) Quartz
- (B) Diamond
- (C) Topaz
- (D) Corundum

Q46. [1-MCQ] [1 mark]

Law of superposition by:

- (A) Steno
- (B) Lyell
- (C) Darwin
- (D) Hutton

Q47. [1-MCQ] [1 mark]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q48. [1-NAT] [1 mark]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q49. [2-NAT] [2 marks]

Igneous rock from lava:

(Numerical Answer Type — Enter the numerical value)

Q50. [2-NAT] [2 marks]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q51. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q52. [2-NAT] [2 marks]

Igneous rock from lava:

(Numerical Answer Type — Enter the numerical value)

Q53. [2-NAT] [2 marks]

Hardest mineral on Mohs scale:

(Numerical Answer Type — Enter the numerical value)

Q54. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q55. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q56. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q57. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q58. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q59. [2-MCQ] [2 marks]

Hardest mineral on Mohs scale:

- (A) Corundum
- (B) Diamond
- (C) Topaz
- (D) Quartz

Q60. [2-MCQ] [2 marks]

Igneous rock from lava:

- (A) Intrusive
- (B) Extrusive
- (C) Metamorphic
- (D) Sedimentary

Q61. [1-MCQ] [1 mark]

Earth's magnetic field generated by:

- (A) Core
- (B) Crust
- (C) Mantle
- (D) Atmosphere

Q62. [1-MCQ] [1 mark]

P-waves are:

- (A) Primary
- (B) Secondary
- (C) Surface
- (D) Love

Q63. [1-MCQ] [1 mark]

Gravity anomaly due to:

- (A) Density variation
- (B) Magnetism
- (C) Electricity
- (D) None

Q64. [1-MCQ] [1 mark]

Landsat uses:

- (A) Optical
- (B) Radar
- (C) Both
- (D) None

Q65. [1-MCQ] [1 mark]

Igneous rock from:

- (A) Sediment
 - (B) Cooling magma
 - (C) Metamorphism
 - (D) Organic
-

ANSWER KEY

Q. No.	Type	Answer	Marks
1	1NAT	B	1
2	1NAT	B	1
3	1MCQ	A	1
4	1MCQ	B	1
5	1MCQ	A	1
6	1MCQ	A	1
7	1MCQ	A	1
8	1MCQ	A	1
9	1MCQ	A	1
10	1MCQ	B	1
11	1MCQ	A	1
12	1MCQ	C	1
13	1MCQ	B	1

14	1NAT	B	1
15	1MCQ	B	1
16	1MCQ	A	1
17	1NAT	B	1
18	1MCQ	A	1
19	1NAT	C	1
20	1NAT	B	1
21	1MCQ	B	1
22	1NAT	A	1
23	1MCQ	C	1
24	1MCQ	A	1
25	1MCQ	C	1
26	1MCQ	B	1
27	1NAT	A	1
28	1NAT	A	1

29	1MCQ	A	1
30	1MCQ	A	1
31	2NAT	A	2
32	2MCQ	B	2
33	2NAT	A	2
34	2MCQ	A	2
35	2MCQ	A	2
36	2NAT	C	2
37	2MSQ	A	2
38	2NAT	B	2
39	2MCQ	C	2
40	2NAT	B	2
41	2MCQ	C	2
42	2MCQ	C	2
43	2MCQ	B	2

44	2NAT	B	2
45	2NAT	A	2
46	2MCQ	A	2
47	2MCQ	A	2
48	2NAT	B	2
49	2MCQ	B	2
50	2MCQ	A	2
51	2MCQ	C	2
52	2MCQ	A	2
53	2MCQ	B	2
54	2MCQ	B	2
55	2NAT	A	2
56	2MCQ	A	2
57	2MCQ	A	2
58	2MCQ	B	2

59	2MCQ	B	2
60	2MCQ	A	2
61	2NAT	C	2
62	2MCQ	C	2
63	2MCQ	B	2
64	2MCQ	B	2
65	2NAT	A	2

DETAILED SOLUTIONS

Q1.

Answer: B

Solution:

Correct Answer: B

Subject: General Aptitude | Topic: Verbal | Difficulty: easy

Q2.

Answer: B

Solution:

Correct Answer: B

Subject: Engineering Mathematics | Topic: Statistics | Difficulty: moderate

Q3.

Answer: A

Solution:

Correct Answer: A

Subject: Geology | Topic: Fossils | Difficulty: moderate

Q4.

Answer: B

Solution:

Correct Answer: B

Subject: Geology | Topic: Minerals | Difficulty: difficult

Q5.

Answer: A

Solution:

Correct Answer: A

Subject: Geology | Topic: Stratigraphy | Difficulty: moderate

Q6.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Magnetism | Difficulty: moderate

Q7.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Seismic | Difficulty: easy

Q8.

Answer: A

Solution:

Correct Answer: A

Subject: Geophysics | Topic: Gravity | Difficulty: easy

Q9.

Answer: A

Solution:

Correct Answer: A

Subject: Remote Sensing & GIS | Topic: Satellite | Difficulty: easy

Q10.

Answer: B

Solution:

Correct Answer: B

Subject: Petrology | Topic: Rocks | Difficulty: easy

Q11.

Answer: A

Solution:

Eigenvalues are 3 and 1. $[1,0]^T$ is eigenvector for eigenvalue 1. $A^{50}[1,0]^T = [1,0]^T$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q12.

Answer: C

Solution:

$\frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1 + 2 = 3$. Hmm. Let me recalc: $\frac{d}{dx} \ln(\sec x) = \tan x$. $\frac{d}{dx} \ln(\tan x) = \sec^2 x / \tan x = 1/(\sin x \cos x) = 2\csc(2x)$. At $\pi/4$: $1 + 2 = 3$. Not in options. The derivative of $\ln(\sec x * \tan x) = \frac{d}{dx}[\ln(\sin x / \cos^2 x)] = \cot x + 2\tan x$. At $\pi/4$: $1+2=3$. Hmm.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q13.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q14.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q15.

Answer: B

Solution:

$(1+1/n)^{(2n)} = [(1+1/n)^n]^2 \rightarrow e^2$ as $n \rightarrow \infty$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q16.

Answer: A

Solution:

$L\{e^{(-at)}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q17.

Answer: B

Solution:

$dy/y = dx/x$. $\ln(y) = \ln(x) + C$. $y = C*x$. Wait: $y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow \ln y = \ln x + C \Rightarrow y = C*x$. That's option A. Hmm, let me check: $y dx - x dy = 0 \Rightarrow y dx = x dy \Rightarrow dy/y = dx/x \Rightarrow y = C*x$. Answer: A.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q18.

Answer: A

Solution:

Standard normal: $P(-1 < Z < 1)$ approx 0.68.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q19.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{(n-2)} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q20.

Answer: B

Solution:

$\text{proj} = ((v \cdot w)/(w \cdot w)) * w = (4/5) * (2, 1, 0) = (8/5, 4/5, 0)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q21.

Answer: B

Solution:

$f(0)=2$, $f(1)=0$, $f(-1)=4$, $f(2)=4$. $f(-2)=-8+6+2=0$. Roots at $x=1$ (double?) and $x=-2$. $f(x)=(x-1)^2(x+2)$. Two distinct real roots.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q22.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q23.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q24.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q25.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q26.

Answer: B

Solution:

$|x|$ is continuous everywhere. Left derivative at 0 = -1, right = +1. Not differentiable at 0.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q27.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4(1+4)} = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q28.

Answer: A

Solution:

$|\langle u, v \rangle| \leq \|u\| * \|v\|$. Equality when u and v are linearly dependent.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q29.

Answer: A

Solution:

Area = $\pi * r^2$. Circumference = $2 * \pi * r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q30.

Answer: A

Solution:

Homogeneous ODE. Substituting $y=x*v$ and $y(1)=0$ gives $y=0$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q31.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q32.

Answer: B

Solution:

$$P(A|B) = P(A \cap B)/P(B) = 0.2/0.4 = 0.5.$$

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q33.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q34.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q35.

Answer: A

Solution:

PGF of Poisson(lambda): $G(z) = E[z^X] = \exp(\lambda(z-1))$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q36.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q37.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q38.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q39.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q40.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q41.

Answer: C

Solution:

Kirchhoff: for K_n , spanning trees = $n^{n-2} = 4^2 = 16$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q42.

Answer: C

Solution:

Gauss-Seidel converges if matrix is diagonally dominant. Jacobi needs symmetric positive definite.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q43.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q44.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q45.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q46.

Answer: A

Solution:

$f(i) = i^2 + 3i + 2 = -1 + 3i$. $\text{Re}(f(i)) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q47.

Answer: A

Solution:

$P(X \geq 2) = 1 - e^{-4}(1+4) = 1 - 5e^{-4} \approx 0.908$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q48.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q49.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q50.

Answer: A

Solution:

Poisson: mean = lambda = variance. Both = 4.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q51.

Answer: C

Solution:

Rolle's: f continuous on [a,b], differentiable on (a,b), $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b).

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q52.

Answer: A

Solution:

$L\{e^{-at}\} = 1/(s+a)$. Here $a=3$: $1/(s+3)$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q53.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: moderate

Q54.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q55.

Answer: A

Solution:

$L\{t^*f(t)\} = -d/ds(F(s))$. $F(s) = 1/(s+2)$. $-d/ds[1/(s+2)] = 1/(s+2)^2$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q56.

Answer: A

Solution:

Row 2 = 2*Row 1, Row 3 = 3*Row 1. All rows linearly dependent. Rank = 1.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q57.

Answer: A

Solution:

$f(x)=x$ is odd on symmetric interval. Fourier series has only sine terms.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: easy

Q58.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q59.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: easy

Q60.

Answer: A

Solution:

Eigenvalues of A^2 are 1, 4, 9. Trace = sum = $1+4+9 = 14$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q61.

Answer: C

Solution:

Rolle's: f continuous on $[a,b]$, differentiable on (a,b) , $f(a)=f(b)$. Then $f'(c)=0$ for some c in (a,b) .

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: difficult

Q62.

Answer: C

Solution:

$y = A\cos(2x) + B\sin(2x)$. $A=1$, $B=0$. $y(\pi/2) = \cos(\pi) = -1$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

Q63.

Answer: B

Solution:

Mohs: Talc(1)..Quartz(7)..Topaz(8)..Corundum(9)..Diamond(10). Diamond hardest.

Subject: Geology | Topic: Geology | Difficulty: easy

Q64.

Answer: B

Solution:

Extrusive (volcanic): from lava at surface. Intrusive: magma below surface.

Subject: Geology | Topic: Geology | Difficulty: difficult

Q65.

Answer: A

Solution:

Area = πr^2 . Circumference = $2\pi r$.

Subject: Engineering Mathematics | Topic: Engineering Mathematics | Difficulty: moderate

PAPER BLUEPRINT

Subject Distribution:

Engineering Mathematics: 42 questions, 65 marks
Geology: 17 questions, 29 marks
Geophysics: 3 questions, 3 marks
General Aptitude: 1 questions, 1 marks
Remote Sensing & GIS: 1 questions, 1 marks
Petrology: 1 questions, 1 marks

Difficulty Distribution:

Easy: 25 | Moderate: 30 | Difficult: 10

Statistics:

Total Questions: 65
Total Marks: 100

This blueprint is generated from the actual question distribution in this paper.